ITEAD SONOFF S31 - SMART PLUG OWNER MANUAL & HARDWARE SPEC

Model: S31 US Type-B | MCU: Espressif ESP8266EX (4MB SPI Flash) | Power IC: CSE7766
FCC ID: 2APN5S31 | Rating: 120V AC 60Hz, Max 15A / 1800W Resistive

1. FACTORY PROPRIETARY FIRMWARE BASELINE (GENESIS BLOCK #0)
   - Shipped Factory Binary: itead/ewelink-oem-firmware v2.6.1 (Closed-Source Cloud Agent)
   - Base SDK: espressif/esp8266-nonos-sdk v2.2.1 | Cloud Broker: us-ota.coolkit.cc:8080
   - Security Note: Factory firmware requires remote eWeLink cloud broker and lacks local API.

2. INTERNAL J1 UART HEADER PINOUT & GPIO MAPPING (FOR ESPHOME LIBERATION FLASH)
   - WARNING: DISCONNECT S31 FROM 120V AC MAINS BEFORE OPENING ENCLOSURE OR CONNECTING UART!
   - J1 Pad 1 (Square): VCC 3.3V DC (Regulated 3.3V only - DO NOT apply 5V!)
   - J1 Pad 2: RX (GPIO3 / UART0 RX - Shared with CSE7766 at 4800 baud EVEN parity)
   - J1 Pad 3: TX (GPIO1 / UART0 TX)
   - J1 Pad 4: GND (Common Ground)
   - Peripherals: GPIO0 = Side Push Button (Active Low / Hold at boot for Flash Mode)
     GPIO12 = SPST 15A Relay (Active High) + Red Status LED
     GPIO13 = Blue Network Status LED (Active Low)

3. CONTINUOUS ATTESTATION & OPEN-SOURCE ESPHOME MIGRATION
   - Initial UART Flash: esphome/esphome v2026.7.0 (Block #1 - 3 versions historical baseline)
   - Upstream Device Profile: https://devices.esphome.io/devices/Sonoff-S31
   - Home Assistant OTA #1: esphome/esphome v2026.8.2 (Block #2 - lwIP Wi-Fi beacon fix)
   - Home Assistant OTA #2: esphome/esphome v2026.9.1 (Block #3 - ESPAsyncWebServer CVE-2026-4108 fix)
   - Home Assistant OTA #3: esphome/esphome v2026.10.3 (Block #4 - Noise API encryption rotation)